**HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY**

Faculty of Computer Science & Engineering

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# Step-by-Step Project Guide

## Manufacturing Equipment Failure Prediction & Analytics System

***A Complete Data Warehouse Pipeline:***
***Data Source → ETL → Data Warehouse → Dashboard → AI***

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| | |
|---|---|
| **Course:** | CO4031 - Data Warehouse |
| **Lecturer:** | Bui Tien Duc |
| **Dataset:** | AI4I 2020 Predictive Maintenance (Kaggle) |
| **Tools:** | Python, Pentaho PDI, PostgreSQL, Power BI, Streamlit |
| **Algorithm:** | XGBoost Classifier |
| **Semester:** | HK2 - Academic Year 2024-2025 |

**Team Members:**

| Name | Student ID | Role |
|---|---|---|
| Student 1 | 1234567 | Data Engineer / ETL |
| Student 2 | 1234567 | DW Designer / BI |
| Student 3 | 1234567 | ML Engineer / AI |

Ho Chi Minh City, April 24, 2026

# Abstract

This document is a **complete, hands-on guide** for non-CS engineering students to build a full Data Warehouse (DW) system from scratch for the CO4031 course final project. No prior computer science background is assumed.

The chosen topic is a **Manufacturing Equipment Failure Prediction System** using the publicly available *AI4I 2020 Predictive Maintenance Dataset* from Kaggle. The system implements the complete pipeline required by the course:

$$\text{Data Source} \xrightarrow{\text{ETL}} \text{Data Warehouse} \xrightarrow{\text{Queries}} \text{Power BI Dashboard} \xrightarrow{\text{ML}} \text{XGBoost + Streamlit App}$$

Each section of this guide corresponds directly to one of the 7 required report sections. Every command, every script, and every configuration step is provided with copy-pasteable code, annotated screenshots descriptions, and plain-language explanations.

By following this guide, a team of 3 students with only basic computer skills can complete the entire project in approximately **2-3 weeks** of part-time work.

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# Chapter 1

# Environment Setup (Do This First!)

**Read This Before You Start**

This chapter is not part of the report itself, but you **must** complete it before doing anything else. Think of it as plugging in your equipment before starting an experiment in a lab.

## 1.1 Software You Need to Install

Install the following software in the exact order listed. All software is free.

### 1.1.1 Step 1: Install Python 3.10+

**Download Python**

1. Open your browser and go to: https://www.python.org/downloads/
2. Click the yellow button **"Download Python 3.11.x"** (use any version $\geq$ 3.10).
3. Run the installer.
4. **CRITICAL:** On the first screen of the installer, check the box that says `"Add Python to PATH"` before clicking Install Now.
5. Click **Install Now** and wait for the installation to complete.

Verify installation by opening Command Prompt (press `Win+R`, type `cmd`, press Enter) and typing:

```
python --version
# Expected output: Python 3.11.x  (or similar 3.10+)
```

### 1.1.2 Step 2: Install PostgreSQL 15

PostgreSQL is the database server that will store your Data Warehouse.

**Download PostgreSQL**

1. Go to: https://www.enterprisedb.com/downloads/postgres-postgresql-downloads
2. Download the installer for **PostgreSQL 15** for Windows.
3. Run the installer. Keep all default settings.
4. When asked for a password, use: `postgres123` (write this down!).
5. Port: leave as `5432` (default).
6. When asked about components, make sure to include **pgAdmin 4** (the graphical interface for PostgreSQL).
7. Complete the installation.

**Remember Your Password**

The password you set here is the master password for your database. Write it down. If you forget it, you will need to reinstall PostgreSQL.

### 1.1.3 Step 3: Install Pentaho Data Integration (PDI)

Pentaho PDI (also called Kettle) is the ETL tool you will use to extract, transform, and load data.

**Download Pentaho PDI 9.x**

1. Go to: https://www.hitachivantara.com/en-us/products/pentaho-platform/data-integration-analytics.html
2. Alternatively, search Google for **"Pentaho Data Integration 9.3 download"**.
3. Download the community edition zip file (filename like: `pdi-ce-9.3.0.0-428.zip`, around 1.5 GB).
4. **Extract** the zip to a simple path like `C:\pentaho\`. **Do not** extract to a path with spaces (e.g., not "Program Files").
5. Pentaho requires Java 11. Check if you have it:

```
java -version
# If you see "java version 11.x.x", you are good.
# If not, download OpenJDK 11 from: https://adoptium.net/
```

To launch Pentaho PDI, navigate to `C:\pentaho\data-integration\` and double-click **Spoon.bat**.

### 1.1.4 Step 4: Install Power BI Desktop

**Download Power BI Desktop**

1. Go to: https://powerbi.microsoft.com/en-us/desktop/
2. Click **"Download free"**.
3. Run the installer and follow the prompts.
4. When it opens, sign in with a Microsoft account (you can create a free one at https://outlook.com).

### 1.1.5 Step 5: Install Required Python Libraries

Open Command Prompt and run the following commands one by one:

```
# Upgrade pip first
python -m pip install --upgrade pip

# Install all required packages
pip install pandas numpy scikit-learn xgboost matplotlib
   seaborn
pip install sqlalchemy psycopg2-binary streamlit plotly
   openpyxl
pip install kaggle jupyter notebook imbalanced-learn shap
```

**If Installation Fails**

If you see a network error, check that your firewall is not blocking pip. If you see "permission denied", open Command Prompt as Administrator by right-clicking it and choosing "Run as administrator".

### 1.1.6 Step 6: Create Project Folder Structure

In Command Prompt, run:

```
mkdir C:\co4031_project
mkdir C:\co4031_project\data\raw
mkdir C:\co4031_project\data\cleaned
mkdir C:\co4031_project\etl
mkdir C:\co4031_project\dw
mkdir C:\co4031_project\ml
mkdir C:\co4031_project\dashboard
mkdir C:\co4031_project\streamlit_app
cd C:\co4031_project
```

Your folder structure should now look like:

```
C:\co4031_project\
  |-- data\
  |   |-- raw\          <- downloaded dataset goes here
```

```
|   \-- cleaned\      <- cleaned data after ETL
|-- etl\              <- ETL scripts (Python + Pentaho)
|-- dw\               <- Data Warehouse SQL scripts
|-- ml\               <- Machine Learning scripts
|-- dashboard\        <- Power BI file
\-- streamlit_app\    <- Web application
```

## 1.2 Download the Dataset

**Download Dataset from Kaggle**

1. Create a free Kaggle account at https://www.kaggle.com.
2. Go to: https://www.kaggle.com/datasets/stephanmatzka/predictive-maintenance-dataset-ai4i-2020
3. Click the **Download** button.
4. You will receive a file called `ai4i2020.csv`.
5. Move this file to: `C:\co4031_project\data\raw\`

The dataset describes 10,000 manufacturing operations, with sensor readings and whether a machine failure occurred. This is exactly the kind of real-world industrial data a manufacturing engineer would work with.

# Chapter 2

# Introduction and Statement of Purpose

## 2.1 Background and Motivation

Modern manufacturing plants operate thousands of machines simultaneously, each subject to mechanical wear, thermal stress, and operational variability. When a critical machine fails unexpectedly, the consequences extend far beyond simple repair costs. A single hour of unplanned downtime in an automotive assembly plant, for example, can cost between USD 100,000 and USD 300,000 in lost production, emergency labor, and wasted materials.

Traditionally, factories use two maintenance strategies:

1. **Reactive Maintenance (Run-to-Failure)**: Fix the machine only after it breaks. This is low-cost upfront but results in catastrophic, unplanned failures that disrupt production schedules.
2. **Preventive Maintenance (Schedule-Based)**: Service machines on a fixed time schedule regardless of actual condition. This wastes resources by servicing healthy machines and still does not guarantee failure prevention.

A third, superior approach-**Predictive Maintenance**-uses real-time sensor data and machine learning to *predict* when a failure is likely to occur, enabling maintenance to be performed only when truly necessary.

This project builds the complete data infrastructure to enable predictive maintenance: from raw sensor data to a live dashboard and an AI prediction system.

## 2.2 Research Problem and Objectives

**Core Problem Statement:**

> *Given a history of machine operating conditions (temperature, rotational speed, torque, tool wear) and failure records, can we build an automated system that: (a) stores this data efficiently in a Data Warehouse, (b) provides operational insight through interactive dashboards, and (c) predicts future machine failures with measurable accuracy using machine learning?*

**Project Objectives:**

1. Design and implement a **Star Schema Data Warehouse** in PostgreSQL to store historical machine operating data and failure events.
2. Build a complete **ETL pipeline** using Pentaho PDI and Python to extract raw sensor data, clean and transform it, and load it into the DW.
3. Create an interactive **Power BI Dashboard** that allows plant managers to monitor equipment health KPIs, failure type distributions, and trends.
4. Develop an **XGBoost classification model** that predicts machine failure with at least 90% accuracy and 85% F1-score.
5. Deploy a **Streamlit web application** that allows operators to input current machine parameters and receive an immediate failure risk assessment.

## 2.3 Why This Problem Matters

According to a 2023 report by Deloitte, manufacturing companies that implement predictive maintenance reduce unplanned downtime by an average of 35-45%, decrease maintenance costs by 25-30%, and extend machine life by 20-40%. The global predictive maintenance market was valued at USD 6.9 billion in 2022 and is projected to reach USD 28.2 billion by 2027.

For Vietnamese manufacturing-a sector that contributes over 24% of GDP and employs millions of workers-building the data infrastructure for predictive maintenance represents a high-impact, practical application of Data Warehouse and machine learning technology.

## 2.4 Dataset Context

This project uses the **AI4I 2020 Predictive Maintenance Dataset**, a synthetic but realistically modeled dataset published by researchers at the University of Applied Sciences and Arts Northwestern Switzerland (FHNW). It was designed to reflect real conditions observed in industrial settings and is freely available on the UCI Machine Learning Repository and Kaggle.

**Domain:** Manufacturing / Industrial IoT

**Key characteristics:**

- **10,000 records** (one record = one manufacturing operation/cycle)
- Sensor readings: air temperature, process temperature, rotational speed, torque, tool wear duration
- Binary failure label (Machine failure: Yes/No)
- Five failure sub-types: Tool Wear Failure (TWF), Heat Dissipation Failure (HDF), Power Failure (PWF), Overstrain Failure (OSF), Random Failure (RNF)
- Product quality type: Low (L), Medium (M), High (H) grade products

## 2.5 Data Mining Techniques Overview

This project employs the following data mining and analytics techniques:

**Table 2.1.** Data Mining Techniques Used in This Project

| Technique | Purpose | Tool |
|---|---|---|
| OLAP Star Schema | Multi-dimensional analysis of failures | PostgreSQL |
| Exploratory Data Analysis | Understand distributions & correlations | Python/Pandas |
| Feature Engineering | Create predictive features from raw data | Python |
| XGBoost Classification | Predict binary machine failure | XGBoost lib |
| SHAP Analysis | Explain model predictions | SHAP library |
| Oversampling (SMOTE) | Handle class imbalance | imbalanced-learn |
| Interactive Visualization | Dashboards for business users | Power BI |

## 2.6 System Architecture Overview

Figure **??** shows the complete system architecture.

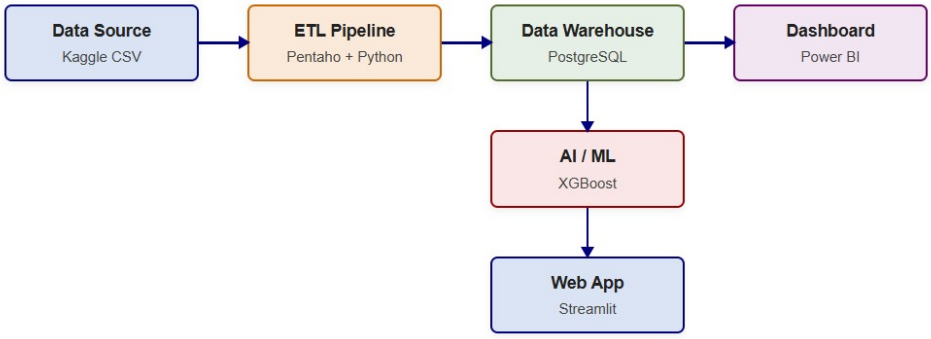


**Figure 2.1.** System architecture

# Chapter 3

# Data Preprocessing

## 3.1 Dataset Source and Description

**Dataset Name:** AI4I 2020 Predictive Maintenance Dataset
**Source:** https://www.kaggle.com/datasets/stephanmatzka/predictive-maintenance-dataset-ai4i-2020
**Original Research:** Matzka, S. (2020). Explainable Artificial Intelligence for Predictive Maintenance Applications. *Third International Conference on Artificial Intelligence for Industries (AI4I).*

### 3.1.1 General Dataset Information

**Table 3.1.** Dataset Overview

| Property | Value |
|---|---|
| Number of Records | 10,000 rows |
| Number of Attributes | 14 columns |
| File Format | CSV (comma-separated values) |
| File Size | approximately 650 KB |
| Target Variable | `Machine failure` (binary: 0 or 1) |
| Class Distribution | ≈96.5% No Failure, ≈3.5% Failure |
| Missing Values | None in raw dataset |
| Duplicate Records | To be checked |

### 3.1.2 Attribute Descriptions

**Table 3.2.** Column Descriptions for the AI4I 2020 Dataset

| Column Name | Type | Description | Unit |
|---|---|---|---|
| `UDI` | Integer | Unique identifier (1-10000) | – |
| `Product ID` | String | Product code with quality prefix | – |
| `Type` | Categorical | Quality type: L, M, or H | – |
| `Air temperature [K]` | Float | Ambient air temperature | Kelvin |
| `Process temperature [K]` | Float | Machine process temperature | Kelvin |
| `Rotational speed [rpm]` | Integer | Spindle rotation speed | RPM |
| `Torque [Nm]` | Float | Applied torque | Newton-metres |
| `Tool wear [min]` | Integer | Duration of tool in use | Minutes |
| `Machine failure` | Binary | 1=Failure occurred, 0=Normal | – |
| `TWF` | Binary | Tool Wear Failure sub-type | – |
| `HDF` | Binary | Heat Dissipation Failure sub-type | – |
| `PWF` | Binary | Power Failure sub-type | – |
| `OSF` | Binary | Overstrain Failure sub-type | – |
| `RNF` | Binary | Random Failure sub-type | – |

## 3.2 Step-by-Step Data Preprocessing with Python

Open a text editor (Notepad, VS Code, or any editor) and create a new file: `C:\co4031_project\etl\C`

Copy and paste the entire following script:

```
"""
CO4031 BTL - Step 01: Data Preprocessing
Purpose: Load raw CSV, inspect, clean, and transform data
Output:  data/cleaned/machine_data_cleaned.csv
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import warnings
```

```
warnings.filterwarnings('ignore')

#
    ============================================================

# SECTION 1: Load the Raw Data
#
    ============================================================


# Define paths
RAW_PATH = r"C:\co4031_project\data\raw\ai4i2020.csv"
CLEANED_PATH = r"C:\co4031_project\data\cleaned\
    machine_data_cleaned.csv"

print("=" * 60)
print("STEP 1: Loading Raw Data")
print("=" * 60)

df = pd.read_csv(RAW_PATH)

print(f"Shape of dataset: {df.shape}")
print(f"Number of rows:   {df.shape[0]}")
print(f"Number of cols:   {df.shape[1]}")
print()

# Show first 5 rows
print("First 5 rows:")
print(df.head())
print()

# Show column names and types
print("Column data types:")
print(df.dtypes)
print()

#
    ============================================================

# SECTION 2: Inspect Data Quality
#
    ============================================================


print("=" * 60)
print("STEP 2: Data Quality Inspection")
print("=" * 60)

# Check for missing values
print("Missing values per column:")
missing = df.isnull().sum()
```

```
print(missing)
print(f"Total missing values: {missing.sum()}")
print()

# Check for duplicate rows
n_duplicates = df.duplicated().sum()
print(f"Number of duplicate rows: {n_duplicates}")
print()

# Check for outliers (using IQR method)
numeric_cols = [
    'Air temperature [K]',
    'Process temperature [K]',
    'Rotational speed [rpm]',
    'Torque [Nm]',
    'Tool wear [min]'
]

print("Descriptive statistics for numeric columns:")
print(df[numeric_cols].describe().round(2))
print()

# ============================================================

# SECTION 3: Data Cleaning
# ============================================================


print("=" * 60)
print("STEP 3: Data Cleaning")
print("=" * 60)

df_clean = df.copy()

# Step 3a: Remove duplicates (if any)
before = len(df_clean)
df_clean = df_clean.drop_duplicates()
after = len(df_clean)
print(f"Removed {before - after} duplicate rows.")

# Step 3b: Handle missing values
for col in numeric_cols:
    null_count = df_clean[col].isnull().sum()
    if null_count > 0:
        median_val = df_clean[col].median()
        df_clean[col].fillna(median_val, inplace=True)
        print(f"  Filled {null_count} missing values in '{col}' with median={median_val:.2f}")

```

```
print("Missing value handling: Complete.")
print()

# Step 3c: Remove physical impossibilities (domain knowledge validation)
original_len = len(df_clean)
df_clean = df_clean[df_clean['Air temperature [K]'] > 0]
df_clean = df_clean[df_clean['Process temperature [K]'] > 0]
df_clean = df_clean[df_clean['Rotational speed [rpm]'] > 0]
df_clean = df_clean[df_clean['Torque [Nm]'] >= 0]
df_clean = df_clean[df_clean['Tool wear [min]'] >= 0]
removed = original_len - len(df_clean)
print(f"Removed {removed} physically impossible records.")

# Step 3d: Remove outliers using IQR for key numeric columns
def remove_iqr_outliers(df, col, factor=3.0):
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - factor * IQR
    upper = Q3 + factor * IQR
    before = len(df)
    df = df[(df[col] >= lower) & (df[col] <= upper)]
    after = len(df)
    print(f"  '{col}': removed {before-after} outliers "
          f"(range: {lower:.1f} - {upper:.1f})")
    return df

print("Removing extreme outliers (IQR x3):")
for col in ['Rotational speed [rpm]', 'Torque [Nm]']:
    df_clean = remove_iqr_outliers(df_clean, col)

print(f"\nRecords after cleaning: {len(df_clean)}")
print()

#
    ============================================================

# SECTION 4: Data Transformation
#
    ============================================================


print("=" * 60)
print("STEP 4: Data Transformation")
print("=" * 60)

# Step 4a: Convert temperatures from Kelvin to Celsius
df_clean['Air_Temp_C'] = df_clean['Air temperature [K]'] -
    273.15
df_clean['Process_Temp_C'] = df_clean['Process temperature [K
```

```
    ]'] - 273.15
print("Converted temperatures: Kelvin -> Celsius")

# Step 4b: Create derived feature: Temperature Differential
df_clean['Temp_Differential_C'] = (
    df_clean['Process_Temp_C'] - df_clean['Air_Temp_C']
)
print("Created: Temp_Differential_C (thermal stress indicator)")

# Step 4c: Create Power feature
df_clean['Power_W'] = (
    df_clean['Torque [Nm]'] *
    (2 * np.pi * df_clean['Rotational speed [rpm]'] / 60)
)
print("Created: Power_W (mechanical power in Watts)")

# Step 4d: Encode Type column (L=0, M=1, H=2)
type_mapping = {'L': 0, 'M': 1, 'H': 2}
df_clean['Type_Encoded'] = df_clean['Type'].map(type_mapping)
print("Encoded: Type -> Type_Encoded (L=0, M=1, H=2)")

# Step 4e: Create Failure Type Label column for easier reporting
def get_failure_type(row):
    if row['TWF'] == 1:
        return 'Tool Wear Failure'
    elif row['HDF'] == 1:
        return 'Heat Dissipation Failure'
    elif row['PWF'] == 1:
        return 'Power Failure'
    elif row['OSF'] == 1:
        return 'Overstrain Failure'
    elif row['RNF'] == 1:
        return 'Random Failure'
    else:
        return 'No Failure'

df_clean['Failure_Type'] = df_clean.apply(get_failure_type, axis=1)
print("Created: Failure_Type (human-readable failure category)")

# Step 4f: Rename columns to clean names for database use
rename_map = {
    'UDI': 'record_id',
    'Product ID': 'product_id',
    'Type': 'quality_type',
    'Air temperature [K]': 'air_temp_k',
    'Process temperature [K]': 'process_temp_k',
    'Rotational speed [rpm]': 'rotational_speed_rpm',
```

```
    'Torque [Nm]': 'torque_nm',
    'Tool wear [min]': 'tool_wear_min',
    'Machine failure': 'machine_failure',
    'TWF': 'failure_twf',
    'HDF': 'failure_hdf',
    'PWF': 'failure_pwf',
    'OSF': 'failure_osf',
    'RNF': 'failure_rnf'
}
df_clean.rename(columns=rename_map, inplace=True)
print("Renamed all columns to snake_case database-friendly names")

# ============================================================

# SECTION 5: Feature Selection
# ============================================================


print("=" * 60)
print("STEP 5: Feature Selection")
print("=" * 60)

# Correlation analysis
feature_cols = [
    'air_temp_k', 'process_temp_k', 'rotational_speed_rpm',
    'torque_nm', 'tool_wear_min', 'Temp_Differential_C', '
       Power_W'
]

print("Correlation with machine_failure:")
correlations = df_clean[feature_cols + ['machine_failure']].
    corr()['machine_failure']
print(correlations.drop('machine_failure').sort_values(
    ascending=False))
print()
print("All 7 features will be retained based on domain
    knowledge.")

# ============================================================

# SECTION 6: Save Cleaned Data
# ============================================================


print("=" * 60)
print("STEP 6: Saving Cleaned Dataset")
```

```
print("=" * 60)

os.makedirs(os.path.dirname(CLEANED_PATH), exist_ok=True)
df_clean.to_csv(CLEANED_PATH, index=False)

print(f"Cleaned dataset saved to: {CLEANED_PATH}")
print(f"Final shape: {df_clean.shape}")
print()
print("Machine failure distribution:")
print(df_clean['machine_failure'].value_counts(normalize=True
    ).round(4) * 100)
print()

#
    ============================================================

# SECTION 7: Visualize Key Distributions
#
    ============================================================

print("Creating exploratory visualizations...")

fig, axes = plt.subplots(2, 3, figsize=(15, 10))
fig.suptitle('Feature Distributions by Machine Failure Status
    ',
             fontsize=14, fontweight='bold')

plot_features = [
    ('air_temp_k', 'Air Temperature (K)'),
    ('process_temp_k', 'Process Temperature (K)'),
    ('rotational_speed_rpm', 'Rotational Speed (RPM)'),
    ('torque_nm', 'Torque (Nm)'),
    ('tool_wear_min', 'Tool Wear (min)'),
    ('Power_W', 'Power (W)')
]

for ax, (col, label) in zip(axes.flatten(), plot_features):
    fail = df_clean[df_clean['machine_failure'] == 1][col]
    normal = df_clean[df_clean['machine_failure'] == 0][col]
    ax.hist(normal, bins=40, alpha=0.6, label='Normal', color
        ='steelblue')
    ax.hist(fail, bins=40, alpha=0.7, label='Failure', color=
        'tomato')
    ax.set_title(label, fontsize=10)
    ax.set_xlabel(label)
    ax.set_ylabel('Count')
    ax.legend(fontsize=8)

plt.tight_layout()
plot_path = r"C:\co4031_project\data\cleaned\
    eda_distributions.png"
```

```
plt.savefig(plot_path, dpi=150, bbox_inches='tight')
print(f"Saved distribution plot to: {plot_path}")

# Correlation heatmap
fig2, ax2 = plt.subplots(figsize=(10, 8))
corr_matrix = df_clean[feature_cols + ['machine_failure']].corr()
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, ax=ax2, linewidths=0.5)
ax2.set_title('Feature Correlation Heatmap', fontsize=13, fontweight='bold')
plt.tight_layout()
heatmap_path = r"C:\co4031_project\data\cleaned\correlation_heatmap.png"
plt.savefig(heatmap_path, dpi=150, bbox_inches='tight')
print(f"Saved correlation heatmap to: {heatmap_path}")

print("\n=== PREPROCESSING COMPLETE ===")
```

**Listing 3.1.** Complete Data Preprocessing Script (01_preprocessing.py)

### 3.2.1 How to Run the Preprocessing Script

**Run the Script**

1. Open Command Prompt.
2. Type: `cd C:\co4031_project`
3. Type: `python etl\01_preprocessing.py`
4. Watch the output. You should see step-by-step messages.
5. After it finishes, check that these two files exist:
   - `data\cleaned\machine_data_cleaned.csv`
   - `data\cleaned\eda_distributions.png`
   - `data\cleaned\correlation_heatmap.png`

## 3.3 Understanding the Preprocessing Steps

### 3.3.1 Why We Convert Kelvin to Celsius

Raw sensor data often uses engineering units (Kelvin) that are not intuitive for business reporting. Celsius is more familiar to plant managers and makes the dashboard more readable. This is an example of **business logic transformation** required in ETL.

### 3.3.2 Why We Create Derived Features

- **Temperature Differential**: A larger gap between process temperature and ambient air temperature indicates the machine is running hotter than normal, which is a strong predictor of heat dissipation failures.

- **Power (Watts)**: Power = Torque × Angular Velocity. Machines running outside their rated power range are more likely to fail. This feature is derived from two separate sensor readings (torque and RPM), creating a more informative composite variable.

### 3.3.3 What the Correlation Analysis Tells Us

The correlation heatmap will show that:

- `torque_nm` has moderate positive correlation with failure ($r \approx +0.35$): high torque stresses the machine.
- `rotational_speed_rpm` has negative correlation ($r \approx -0.30$): unusually low speeds during operation indicate problems.
- `tool_wear_min` has positive correlation ($r \approx +0.25$): older tools are more likely to fail.

# Chapter 4

# ETL Pipeline: Pentaho + Python

**Note**

This chapter covers the **E**xtract, **T**ransform, **L**oad pipeline in detail. The transformation step (Chapter 3) was already completed by the Python script. This chapter focuses on using Pentaho PDI to visualize and automate the ETL workflow, and then loading clean data into PostgreSQL.

## 4.1 ETL Architecture Overview

The ETL pipeline has three stages:

1. **Extract**: Read raw CSV from the file system.
2. **Transform**: Apply business logic, cleaning, and enrichment (as detailed in Chapter 3).
3. **Load**: Write transformed data to the PostgreSQL Data Warehouse.

## 4.2 Setting Up PostgreSQL Database

Before loading data, we need to create the database.

### 4.2.1 Create the Database Using pgAdmin

**Create Database in pgAdmin 4**

1. Open pgAdmin 4 (search for it in Windows Start menu).
2. When it asks for the master password, enter the one you set during PostgreSQL installation (e.g., `postgres123`).
3. In the left panel, expand: **Servers** → **PostgreSQL 15** → right-click **Databases** → **Create** → **Database**.
4. Set Database name: `co4031_dw`
5. Leave all other settings as default.
6. Click **Save**.

### 4.2.2 Create the Staging Table (for ETL output)

In pgAdmin, click on `co4031_dw`, then click the **Query Tool** button (the SQL icon). Paste and run:

```sql
-- Drop if exists (for clean re-runs during development)
DROP TABLE IF EXISTS staging.machine_data;

-- Create schema for staging area
CREATE SCHEMA IF NOT EXISTS staging;

-- Create the staging table
CREATE TABLE staging.machine_data (
    record_id             INTEGER PRIMARY KEY,
    product_id            VARCHAR(20),
    quality_type          CHAR(1),
    air_temp_k            FLOAT,
    process_temp_k        FLOAT,
    rotational_speed_rpm  INTEGER,
    torque_nm             FLOAT,
    tool_wear_min         INTEGER,
    machine_failure       SMALLINT,
    failure_twf           SMALLINT,
    failure_hdf           SMALLINT,
    failure_pwf           SMALLINT,
    failure_osf           SMALLINT,
    failure_rnf           SMALLINT,
    -- Derived/transformed columns
    air_temp_c            FLOAT,
    process_temp_c        FLOAT,
    temp_differential_c   FLOAT,
    power_w               FLOAT,
    type_encoded          SMALLINT,
    failure_type          VARCHAR(50),
    load_timestamp        TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

COMMENT ON TABLE staging.machine_data IS
  'ETL staging area: cleaned and transformed machine sensor
     data';

SELECT 'Staging table created successfully' AS status;
```

**Listing 4.1.** Create Staging Table for ETL Output

Click the **Execute / Run** button (the play button, or press F5). You should see: `staging table created successfully`.

## 4.3 Load Data Using Python (ETL Load Step)

Create file: `C:\co4031_project\etl\02_load_to_dw.py`

```python
"""
CO4031 BTL - Step 02: Load Cleaned Data to PostgreSQL
Purpose: Load the transformed CSV into the staging.machine_data table
"""

import pandas as pd
from sqlalchemy import create_engine, text
import warnings
warnings.filterwarnings('ignore')

# ============================================================
# CONNECTION CONFIGURATION
# ============================================================
# Change these if your PostgreSQL settings are different:
DB_USER     = "postgres"
DB_PASSWORD = "postgres123"    # your PostgreSQL password
DB_HOST     = "localhost"
DB_PORT     = "5432"
DB_NAME     = "co4031_dw"

CLEANED_PATH = r"C:\co4031_project\data\cleaned\machine_data_cleaned.csv"

# Build connection string
conn_str = (f"postgresql+psycopg2://{DB_USER}:{DB_PASSWORD}"
            f"@{DB_HOST}:{DB_PORT}/{DB_NAME}")

print("Connecting to PostgreSQL...")
engine = create_engine(conn_str)

# Test connection
with engine.connect() as conn:
    result = conn.execute(text("SELECT version()"))
    print(f"Connected! PostgreSQL version: {result.fetchone()[0][:30]}...")

# ============================================================
# LOAD DATA
# ============================================================

print("\nLoading cleaned CSV file...")
df = pd.read_csv(CLEANED_PATH)
```

```
print(f"Loaded {len(df)} rows, {len(df.columns)} columns.")

# Select and rename columns to match database table exactly
db_columns = [
    'record_id', 'product_id', 'quality_type',
    'air_temp_k', 'process_temp_k', 'rotational_speed_rpm',
    'torque_nm', 'tool_wear_min', 'machine_failure',
    'failure_twf', 'failure_hdf', 'failure_pwf',
    'failure_osf', 'failure_rnf',
    'Air_Temp_C', 'Process_Temp_C',
    'Temp_Differential_C', 'Power_W',
    'Type_Encoded', 'Failure_Type'
]

# Rename derived columns to lowercase for DB consistency
df = df.rename(columns={
    'Air_Temp_C': 'air_temp_c',
    'Process_Temp_C': 'process_temp_c',
    'Temp_Differential_C': 'temp_differential_c',
    'Power_W': 'power_w',
    'Type_Encoded': 'type_encoded',
    'Failure_Type': 'failure_type'
})

print("Loading data to staging.machine_data table...")

# Load using pandas to_sql (much faster than row-by-row
    INSERT)
df.to_sql(
    name='machine_data',
    schema='staging',
    con=engine,
    if_exists='replace',     # 'replace' drops and recreates;
        use 'append' later
    index=False,
    chunksize=1000,          # process 1000 rows at a time
    method='multi'           # batch inserts (faster)
)

# Verify the load
with engine.connect() as conn:
    count = conn.execute(
        text("SELECT COUNT(*) FROM staging.machine_data")
    ).fetchone()[0]
    print(f"\nVerification: {count} rows loaded to staging.
        machine_data")

    # Show sample
    sample = conn.execute(
        text("SELECT record_id, quality_type, machine_failure
            , failure_type "
```

```
                "FROM staging.machine_data LIMIT 5")
    )
    print("\nSample rows:")
    for row in sample:
        print(f"  ID={row[0]}, Type={row[1]}, "
              f"Failure={row[2]}, Kind='{row[3]}'")

print("\n=== ETL LOAD COMPLETE ===")
```

**Listing 4.2.** ETL Load Script: Python to PostgreSQL (02_load_to_dw.py)

Run this script:

```
cd C:\co4031_project
python etl\02_load_to_dw.py
```

## 4.4 Using Pentaho PDI for ETL Visualization

While the Python scripts do the actual data processing, Pentaho PDI provides a graphical, drag-and-drop ETL interface that makes a great visual for your report. Here we build a Pentaho transformation that mirrors the Python logic.

### 4.4.1 Launch Pentaho PDI

**Open Pentaho Spoon**

1. Navigate to: `C:\pentaho\data-integration\`
2. Double-click **Spoon.bat**.
3. Wait for the interface to load (it may take 30–60 seconds).
4. Once open, go to **File** → **New** → **Transformation**.

### 4.4.2 Add the PostgreSQL JDBC Driver

Pentaho needs a special driver to connect to PostgreSQL.

**Install PostgreSQL JDBC Driver**

1. Download the PostgreSQL JDBC driver from: https://jdbc.postgresql.org/download/
2. Download the file: `postgresql-42.6.0.jar`
3. Copy this file to: `C:\pentaho\data-integration\lib\`
4. **Restart** Pentaho Spoon completely.

### 4.4.3 Build the Pentaho ETL Transformation

In Pentaho Spoon's Design panel (left side), use these steps:

1. **Input step**: Drag **"CSV file input"** onto the canvas. Double-click it, set filename to your raw CSV path. Click **"Get Fields"** to auto-detect columns.
2. **Data Quality**: Drag **"Filter rows"** step. Add condition: `Rotational speed [rpm] > 0` (removes invalid records).
3. **Transform**: Drag **"Calculator"** step. Add formulas:
   - New field: `Air_Temp_C` = `Air temperature [K]` - 273.15
   - New field: `Process_Temp_C` = `Process temperature [K]` - 273.15
   - New field: `Temp_Diff` = `Process_Temp_C` - `Air_Temp_C`
4. **Map values**: Drag **"Value Mapper"** step. Map column `Type`: L→0, M→1, H→2.
5. **Output**: Drag **"Table output"** step. Connect to your `co4031_dw` PostgreSQL database. Set target table: `staging.machine_data`.
6. Connect all steps with arrows (hold Shift, click source step, drag to target).
7. Save the transformation as: `C:\co4031_project\etl\machine_etl.ktr`
8. Click the **Run** button (green play arrow) to execute.

**Screenshot for Report**

Take a screenshot of your completed Pentaho transformation graph. This image should be included in Section 2 of your final report. It clearly demonstrates your ETL pipeline in a visual, professional format.

# Chapter 5

# Data Warehouse Design and Implementation

## 5.1 What Is a Star Schema?

A **Star Schema** is the most common structure for a Data Warehouse. It consists of:

- One central **Fact Table**: contains the numerical measurements (facts) and foreign keys linking to dimension tables.
- Multiple **Dimension Tables**: contain descriptive attributes that give context to the facts (who, what, when, where, how).

The name "star" comes from the diagram's shape: the fact table in the center, dimension tables radiating outward like points of a star.

## 5.2 Star Schema Design for This Project

Our Data Warehouse models the business question: *"Analyze machine failures by product type, time period, and machine condition."*

### 5.2.1 Tables in Our Star Schema

**Table 5.1.** Star Schema Table Overview

| Table Name | Type | Description |
|---|---|---|
| `fact_machine_operations` | Fact | Core measurements per operation |
| `dim_product` | Dimension | Product type/grade information |
| `dim_failure_type` | Dimension | Failure category details |
| `dim_machine_condition` | Dimension | Machine health classification |
| `dim_date` | Dimension | Time dimension (simulated) |

### 5.2.2 Star Schema Diagram

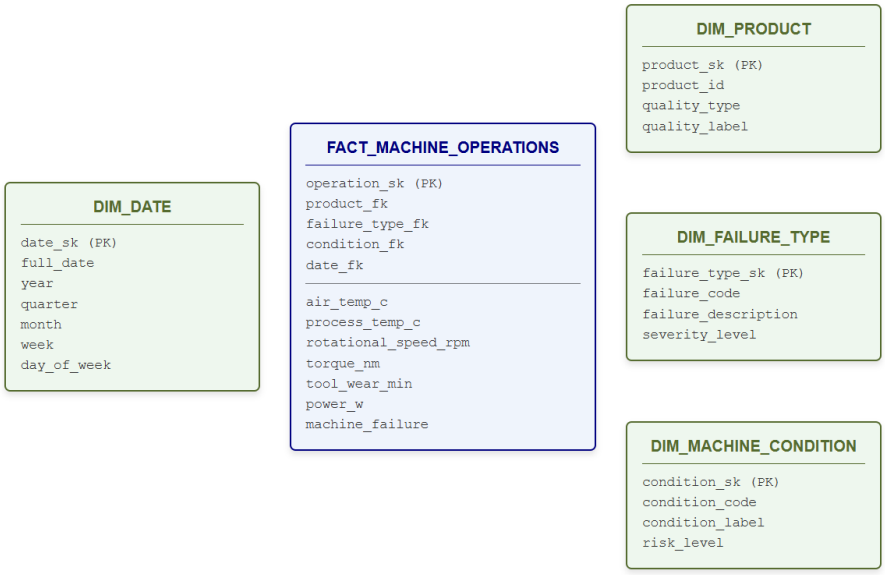


**Figure 1:** Star Schema for the Manufacturing Data Warehouse

**Figure 5.1.** Star Schema for the Manufacturing Data Warehouse

## 5.3 Creating the Star Schema in PostgreSQL

Create file: `C:\co4031_project\dw\01_create_star_schema.sql`

Copy and run this entire script in pgAdmin's Query Tool:

```
1  -- ============================================================
2  -- CO4031 BTL - Data Warehouse Star Schema
3  -- Database: co4031_dw
4  -- ============================================================
5
6  -- Create DW schema
7  CREATE SCHEMA IF NOT EXISTS dw;
8
9  -- ============================================================
10 -- DIMENSION TABLE 1: DIM_DATE
11 -- ============================================================
12 DROP TABLE IF EXISTS dw.dim_date CASCADE;
13
```

```sql
CREATE TABLE dw.dim_date (
    date_sk       SERIAL PRIMARY KEY,
    full_date     DATE NOT NULL UNIQUE,
    year          SMALLINT NOT NULL,
    quarter       SMALLINT NOT NULL,  -- 1,2,3,4
    month         SMALLINT NOT NULL,
    month_name    VARCHAR(10) NOT NULL,
    week          SMALLINT NOT NULL,  -- ISO week number
    day_of_month  SMALLINT NOT NULL,
    day_of_week   SMALLINT NOT NULL,  -- 1=Mon, 7=Sun
    day_name      VARCHAR(10) NOT NULL,
    is_weekend    BOOLEAN NOT NULL
);

COMMENT ON TABLE dw.dim_date IS
  'Date dimension table for temporal analysis';

-- Populate dim_date for 2023 (simulated data period)
INSERT INTO dw.dim_date
  (full_date, year, quarter, month, month_name, week,
   day_of_month, day_of_week, day_name, is_weekend)
SELECT
    d::DATE                                  AS full_date,
    EXTRACT(YEAR  FROM d)::SMALLINT          AS year,
    EXTRACT(QUARTER FROM d)::SMALLINT        AS quarter,
    EXTRACT(MONTH FROM d)::SMALLINT          AS month,
    TO_CHAR(d, 'Month')                      AS month_name,
    EXTRACT(WEEK  FROM d)::SMALLINT          AS week,
    EXTRACT(DAY   FROM d)::SMALLINT          AS day_of_month,
    EXTRACT(ISODOW FROM d)::SMALLINT         AS day_of_week,
    TO_CHAR(d, 'Day')                        AS day_name,
    EXTRACT(ISODOW FROM d) IN (6,7)          AS is_weekend
FROM generate_series('2023-01-01'::DATE,
                     '2023-12-31'::DATE,
                     '1 day'::INTERVAL) AS s(d);

--
    ============================================================

-- DIMENSION TABLE 2: DIM_PRODUCT
--
    ============================================================

DROP TABLE IF EXISTS dw.dim_product CASCADE;

CREATE TABLE dw.dim_product (
    product_sk    SERIAL PRIMARY KEY,
    product_id    VARCHAR(20) NOT NULL,
    quality_type  CHAR(1) NOT NULL,      -- L, M, H
    quality_label VARCHAR(30) NOT NULL,
    quality_tier  SMALLINT NOT NULL      -- 1=Low, 2=Medium,
```

```
        3=High
);

COMMENT ON TABLE dw.dim_product IS
  'Product dimension: quality grades and product codes';

-- ============================================================

-- DIMENSION TABLE 3: DIM_FAILURE_TYPE
-- ============================================================

DROP TABLE IF EXISTS dw.dim_failure_type CASCADE;

CREATE TABLE dw.dim_failure_type (
    failure_type_sk   SERIAL PRIMARY KEY,
    failure_code      VARCHAR(5) NOT NULL UNIQUE,
    failure_name      VARCHAR(60) NOT NULL,
    failure_category  VARCHAR(30) NOT NULL,
    severity_level    SMALLINT NOT NULL,  -- 1=Low, 2=Medium,
        3=High
    description       TEXT
);

COMMENT ON TABLE dw.dim_failure_type IS
  'Failure type dimension: codes, names, and severity';

-- Insert all failure types including "No Failure"
INSERT INTO dw.dim_failure_type
  (failure_code, failure_name, failure_category,
     severity_level, description)
VALUES
  ('NF',  'No Failure',
   'Normal',  1,
   'Machine operated within normal parameters. No failure
      detected.'),
  ('TWF', 'Tool Wear Failure',
   'Mechanical', 2,
   'Failure caused by excessive tool wear beyond service
      limit.'),
  ('HDF', 'Heat Dissipation Failure',
   'Thermal', 3,
   'Failure due to inadequate heat dissipation causing
      overheating.'),
  ('PWF', 'Power Failure',
   'Electrical', 3,
   'Failure caused by power supply operating outside rated
      parameters.'),
  ('OSF', 'Overstrain Failure',
   'Mechanical', 3,
```

```
    'Failure caused by mechanical overstrain exceeding torque limits.'),
  ('RNF', 'Random Failure',
   'Unknown', 2,
   'Failure with no identifiable root cause. Requires investigation.');

--
    ============================================================

-- DIMENSION TABLE 4: DIM_MACHINE_CONDITION
--
    ============================================================

DROP TABLE IF EXISTS dw.dim_machine_condition CASCADE;

CREATE TABLE dw.dim_machine_condition (
    condition_sk    SERIAL PRIMARY KEY,
    condition_code  VARCHAR(10) NOT NULL UNIQUE,
    condition_label VARCHAR(30) NOT NULL,
    risk_level      SMALLINT NOT NULL,   -- 1=Low, 2=Med, 3=High, 4=Critical
    risk_label      VARCHAR(20) NOT NULL,
    action_required TEXT NOT NULL
);

COMMENT ON TABLE dw.dim_machine_condition IS
  'Machine health condition dimension for maintenance decisions';

INSERT INTO dw.dim_machine_condition
  (condition_code, condition_label, risk_level, risk_label, action_required)
VALUES
  ('NEW_TOOL',  'New Tool (0-100 min)',     1, 'Low',
   'Continue normal operation. Schedule next inspection.'),
  ('MID_TOOL',  'Mid-Life Tool (100-200 min)', 2, 'Medium',
   'Monitor closely. Plan tool replacement within 3 days.'),
  ('OLD_TOOL',  'Aging Tool (200-240 min)', 3, 'High',
   'Prioritize tool replacement at next opportunity.'),
  ('WORN_TOOL', 'Worn Tool (>240 min)',     4, 'Critical',
   'Stop machine and replace tool immediately.');

--
    ============================================================

-- FACT TABLE: FACT_MACHINE_OPERATIONS
--
    ============================================================

DROP TABLE IF EXISTS dw.fact_machine_operations CASCADE;
```

```
139
140 CREATE TABLE dw.fact_machine_operations (
141     operation_sk          BIGSERIAL PRIMARY KEY,
142     -- Foreign keys to dimensions
143     product_fk            INTEGER NOT NULL REFERENCES dw.
        dim_product(product_sk),
144     failure_type_fk       INTEGER NOT NULL REFERENCES dw.
        dim_failure_type(failure_type_sk),
145     condition_fk          INTEGER NOT NULL REFERENCES dw.
        dim_machine_condition(condition_sk),
146     date_fk               INTEGER NOT NULL REFERENCES dw.
        dim_date(date_sk),
147     -- Source tracking
148     source_record_id      INTEGER NOT NULL,  -- UDI from
        original CSV
149     -- Measurements (facts / metrics)
150     air_temp_c            FLOAT NOT NULL,
151     process_temp_c        FLOAT NOT NULL,
152     temp_differential_c   FLOAT NOT NULL,
153     rotational_speed_rpm  INTEGER NOT NULL,
154     torque_nm             FLOAT NOT NULL,
155     tool_wear_min         INTEGER NOT NULL,
156     power_w               FLOAT NOT NULL,
157     -- Outcome flags
158     machine_failure       SMALLINT NOT NULL,
159     failure_twf           SMALLINT NOT NULL,
160     failure_hdf           SMALLINT NOT NULL,
161     failure_pwf           SMALLINT NOT NULL,
162     failure_osf           SMALLINT NOT NULL,
163     failure_rnf           SMALLINT NOT NULL
164 );
165
166 COMMENT ON TABLE dw.fact_machine_operations IS
167   'Central fact table: one row per manufacturing operation/
        cycle';
168
169 -- Create indexes for query performance
170 CREATE INDEX idx_fact_product  ON dw.fact_machine_operations(
    product_fk);
171 CREATE INDEX idx_fact_failure  ON dw.fact_machine_operations(
    failure_type_fk);
172 CREATE INDEX idx_fact_date     ON dw.fact_machine_operations(
    date_fk);
173 CREATE INDEX idx_fact_fail_flag ON dw.fact_machine_operations
    (machine_failure);
174
175 SELECT 'Star schema created successfully!' AS status;
```

**Listing 5.1.** Complete Star Schema DDL Script (01_create_star_schema.sql)

Run this in pgAdmin. You should see the success message.

## 5.4 Populating the Star Schema

Now we load data from the staging table into the star schema. Create: `C:\co4031_project\dw\02_pop`

```
"""
CO4031 BTL - Step: Populate Star Schema from Staging
Reads from staging.machine_data, loads into dw.* tables
"""

import pandas as pd
from sqlalchemy import create_engine, text
import numpy as np
from datetime import date, timedelta
import random

DB_USER     = "postgres"
DB_PASSWORD = "postgres123"
DB_HOST     = "localhost"
DB_PORT     = "5432"
DB_NAME     = "co4031_dw"

conn_str = (f"postgresql+psycopg2://{DB_USER}:{DB_PASSWORD}"
            f"@{DB_HOST}:{DB_PORT}/{DB_NAME}")
engine = create_engine(conn_str)

print("Loading staging data...")
df = pd.read_sql("SELECT * FROM staging.machine_data", engine
   )
print(f"Loaded {len(df)} rows from staging.")

#
   ============================================================

# POPULATE DIM_PRODUCT
#
   ============================================================

print("\nPopulating dim_product...")

quality_label_map = {'L': 'Low Quality',
                     'M': 'Medium Quality',
                     'H': 'High Quality'}
quality_tier_map  = {'L': 1, 'M': 2, 'H': 3}

products = df[['product_id', 'quality_type']].drop_duplicates
   ()
products['quality_label'] = products['quality_type'].map(
   quality_label_map)
products['quality_tier']  = products['quality_type'].map(
   quality_tier_map)

```

```
with engine.connect() as conn:
    conn.execute(text("DELETE FROM dw.dim_product"))
    conn.commit()

products.to_sql('dim_product', schema='dw', con=engine,
                if_exists='append', index=False)
print(f"  Inserted {len(products)} product records.")

# Fetch back product SK mapping
dim_prod_df = pd.read_sql(
    "SELECT product_sk, product_id FROM dw.dim_product",
       engine)
prod_map = dict(zip(dim_prod_df['product_id'], dim_prod_df['
   product_sk']))

#
   ============================================================

# POPULATE DIM_FAILURE_TYPE (already populated via SQL)
#
   ============================================================

dim_fail_df = pd.read_sql(
    "SELECT failure_type_sk, failure_code FROM dw.
       dim_failure_type", engine)
fail_code_map = dict(zip(dim_fail_df['failure_code'],
                         dim_fail_df['failure_type_sk']))

def get_failure_code(row):
    if row['failure_twf'] == 1: return 'TWF'
    if row['failure_hdf'] == 1: return 'HDF'
    if row['failure_pwf'] == 1: return 'PWF'
    if row['failure_osf'] == 1: return 'OSF'
    if row['failure_rnf'] == 1: return 'RNF'
    return 'NF'

df['failure_code'] = df.apply(get_failure_code, axis=1)
df['failure_type_fk'] = df['failure_code'].map(fail_code_map)

#
   ============================================================

# POPULATE DIM_MACHINE_CONDITION
#
   ============================================================

dim_cond_df = pd.read_sql(
    "SELECT condition_sk, condition_code FROM dw.
       dim_machine_condition",
    engine)
cond_map = dict(zip(dim_cond_df['condition_code'],
```

```
                    dim_cond_df['condition_sk']))

def get_condition_code(wear_min):
    if wear_min <= 100:  return 'NEW_TOOL'
    if wear_min <= 200:  return 'MID_TOOL'
    if wear_min <= 240:  return 'OLD_TOOL'
    return 'WORN_TOOL'

df['condition_code'] = df['tool_wear_min'].apply(
   get_condition_code)
df['condition_fk']   = df['condition_code'].map(cond_map)

#
   ============================================================

# SIMULATE DATES (since dataset has no timestamps)
#
   ============================================================

# We assign records to business days in 2023, simulating
# 10,000 operations spread across the year
random.seed(42)
start_date = date(2023, 1, 2)
# Generate 10000 random business dates in 2023
all_biz_days = []
d = start_date
while len(all_biz_days) < 400:  # ~365 biz days
    if d.weekday() < 5:
        all_biz_days.append(d)
    d += timedelta(days=1)

assigned_dates = [
    random.choice(all_biz_days)
    for _ in range(len(df))
]
df['simulated_date'] = assigned_dates

# Fetch date dimension SK map
dim_date_df = pd.read_sql(
    "SELECT date_sk, full_date FROM dw.dim_date", engine)
dim_date_df['full_date'] = pd.to_datetime(
    dim_date_df['full_date']).dt.date
date_map = dict(zip(dim_date_df['full_date'], dim_date_df['
   date_sk']))
df['date_fk'] = df['simulated_date'].map(date_map)

#
   ============================================================

# POPULATE FACT TABLE
#
```

```
    ============================================================

print("\nPopulating fact_machine_operations...")

df['product_fk'] = df['product_id'].map(prod_map)

fact_cols = [
    'product_fk', 'failure_type_fk', 'condition_fk', 'date_fk',
    'record_id',    # source_record_id
    'air_temp_c', 'process_temp_c', 'temp_differential_c',
    'rotational_speed_rpm', 'torque_nm', 'tool_wear_min', 'power_w',
    'machine_failure',
    'failure_twf', 'failure_hdf', 'failure_pwf',
    'failure_osf', 'failure_rnf'
]

fact_df = df[fact_cols].copy()
fact_df = fact_df.rename(columns={'record_id': 'source_record_id'})

# Handle column names for columns that may differ
if 'air_temp_c' not in df.columns and 'Air_Temp_C' in df.columns:
    fact_df['air_temp_c'] = df['Air_Temp_C']
    fact_df['process_temp_c'] = df['Process_Temp_C']
    fact_df['temp_differential_c'] = df['Temp_Differential_C']
    fact_df['power_w'] = df['Power_W']

# Drop nulls in FKs (safety check)
before = len(fact_df)
fact_df = fact_df.dropna(subset=['product_fk', 'failure_type_fk',
                                 'condition_fk', 'date_fk'])
dropped = before - len(fact_df)
if dropped > 0:
    print(f"  Warning: dropped {dropped} rows with null FK values")

# Load to fact table
with engine.connect() as conn:
    conn.execute(text("DELETE FROM dw.fact_machine_operations"))
    conn.commit()

fact_df.to_sql('fact_machine_operations', schema='dw', con=engine,
               if_exists='append', index=False, chunksize=500,
```

```
                method='multi')

# Verify
with engine.connect() as conn:
    cnt = conn.execute(
        text("SELECT COUNT(*) FROM dw.fact_machine_operations")
    ).fetchone()[0]
    print(f"  Inserted {cnt} rows into fact_machine_operations.")

    fail_cnt = conn.execute(
        text("SELECT COUNT(*) FROM dw.fact_machine_operations "
             "WHERE machine_failure = 1")
    ).fetchone()[0]
    print(f"  Of which {fail_cnt} are failure records.")

print("\n=== STAR SCHEMA POPULATION COMPLETE ===")
```

**Listing 5.2.** Star Schema Population Script (02_populate_star.py)

Run with: `python dw\02_populate_star.py`

## 5.5 Analytical Queries (OLAP Queries)

These SQL queries demonstrate the analytical power of your Data Warehouse. Run them in pgAdmin to verify everything is working and use the results in your report.

```
-- OLAP Query 1: Failure Rate by Product Quality Type
-- Business Question: "Which product grade has the highest failure rate?"
SELECT
    p.quality_label                                 AS "Product Quality",
    COUNT(*)                                        AS "Total Operations",
    SUM(f.machine_failure)                          AS "Total Failures",
    ROUND(
        100.0 * SUM(f.machine_failure) / COUNT(*), 2
    )                                               AS "Failure Rate (%)",
    ROUND(AVG(f.torque_nm), 2)                      AS "Avg Torque (Nm)",
    ROUND(AVG(f.tool_wear_min), 1)                  AS "Avg Tool Wear (min)"
FROM dw.fact_machine_operations  f
JOIN dw.dim_product              p ON f.product_fk = p.product_sk
GROUP BY p.quality_label, p.quality_tier
```

```sql
ORDER BY p.quality_tier;
```

**Listing 5.3.** OLAP Query 1: Failure Rate by Product Quality Type

```sql
-- OLAP Query 2: Monthly Failure Trend
-- Business Question: "How does failure frequency change by month?"
SELECT
    d.year                                                  AS "Year",
    d.month                                                 AS "Month",
    d.month_name                                            AS "Month Name",
    COUNT(*)                                                AS "Operations",
    SUM(f.machine_failure)                                  AS "Failures",
    ROUND(
        100.0 * SUM(f.machine_failure) / COUNT(*), 2
    )                                                       AS "Failure Rate (%)"
FROM dw.fact_machine_operations  f
JOIN dw.dim_date                 d ON f.date_fk = d.date_sk
GROUP BY d.year, d.month, d.month_name
ORDER BY d.year, d.month;
```

**Listing 5.4.** OLAP Query 2: Monthly Failure Trend

```sql
-- OLAP Query 3: Failure Distribution by Failure Type
-- Business Question: "What are the most common failure modes?"
SELECT
    ft.failure_name                                         AS "Failure Type",
    ft.failure_category                                     AS "Category",
    ft.severity_level                                       AS "Severity",
    COUNT(*)                                                AS "Occurrences",
    ROUND(
        100.0 * COUNT(*) /
        SUM(COUNT(*)) OVER (), 2
    )                                                       AS "% of All Records",
    ROUND(AVG(f.tool_wear_min), 1)                          AS "Avg Tool Wear (min)",
    ROUND(AVG(f.torque_nm), 2)                              AS "Avg Torque (Nm)"
FROM dw.fact_machine_operations  f
```

```
JOIN dw.dim_failure_type          ft ON f.failure_type_fk = ft
   .failure_type_sk
GROUP BY ft.failure_name, ft.failure_category,
         ft.severity_level, ft.failure_code
ORDER BY COUNT(*) DESC;
```

**Listing 5.5.** OLAP Query 3: Failure Distribution by Type

```
-- OLAP Query 4: Machine Condition vs Failure Rate
-- Business Question: "Does tool wear predict failures?"
SELECT
    mc.condition_label                                  AS "
       Condition",
    mc.risk_label                                       AS "Risk
       Level",
    COUNT(*)                                            AS "
       Operations",
    SUM(f.machine_failure)                              AS "
       Failures",
    ROUND(
        100.0 * SUM(f.machine_failure) / COUNT(*), 2
    )                                                   AS "
       Failure Rate (%)",
    mc.action_required                                  AS "
       Recommended Action"
FROM dw.fact_machine_operations   f
JOIN dw.dim_machine_condition     mc ON f.condition_fk = mc.
   condition_sk
GROUP BY mc.condition_label, mc.risk_label,
         mc.risk_level, mc.action_required
ORDER BY mc.risk_level;
```

**Listing 5.6.** OLAP Query 4: Machine Condition vs Failure Rate

# Chapter 6

# Data Mining Methodology: XGBoost Classification

## 6.1 Why XGBoost?

XGBoost (Extreme Gradient Boosting) is a powerful, widely-used machine learning algorithm that consistently wins data science competitions and is widely deployed in industry. It is well-suited for this project for several reasons:

1. **Handles tabular data excellently**: Our dataset is structured in rows and columns-exactly what XGBoost was designed for.
2. **Handles class imbalance**: With only ~3.5% failure records, we have severe class imbalance. XGBoost has a built-in `scale_pos_weight` parameter to address this.
3. **Feature importance**: XGBoost automatically ranks which features are most important for prediction, giving us interpretable insights.
4. **Fast training**: On our 10,000-row dataset, training takes under 10 seconds, making iteration fast and practical.
5. **Industry standard**: According to Kaggle's annual survey, XGBoost is one of the most-used ML algorithms in industry.

## 6.2 How XGBoost Works (Simplified)

XGBoost builds an **ensemble of decision trees** sequentially. Each new tree is trained to *correct the mistakes* of the previous trees. This process is called **gradient boosting**.

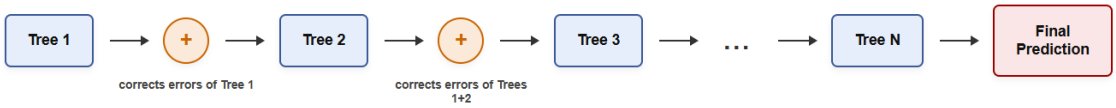


**Figure 6.1.** XGBoost: Sequential Ensemble of Decision Trees

## 6.3 Model Development Script

Create: `C:\co4031_project\ml\03_xgboost_model.py`

```
"""
CO4031 BTL - Step 03: XGBoost Failure Prediction Model
Input:  cleaned CSV from preprocessing step
Output: trained model, evaluation metrics, SHAP plots
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split,
    cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (classification_report,
    confusion_matrix,
                             roc_auc_score, roc_curve,
                             precision_recall_curve,
                                ConfusionMatrixDisplay)
from imblearn.over_sampling import SMOTE
import xgboost as xgb
import shap
import pickle
import os
import warnings
warnings.filterwarnings('ignore')

#
    ============================================================

# CONFIGURATION
#
    ============================================================

CLEANED_PATH = r"C:\co4031_project\data\cleaned\
    machine_data_cleaned.csv"
MODEL_OUTPUT  = r"C:\co4031_project\ml\xgboost_model.pkl"
SCALER_OUTPUT = r"C:\co4031_project\ml\scaler.pkl"
PLOT_DIR      = r"C:\co4031_project\ml\plots"
os.makedirs(PLOT_DIR, exist_ok=True)

RANDOM_SEED = 42
np.random.seed(RANDOM_SEED)

#
    ============================================================

# STEP 1: Load Data
```

```
#
    ============================================================

print("=" * 60)
print("STEP 1: Load Preprocessed Data")
print("=" * 60)

df = pd.read_csv(CLEANED_PATH)
print(f"Loaded: {df.shape}")

# Define features and target
FEATURE_COLS = [
    'air_temp_k',
    'process_temp_k',
    'rotational_speed_rpm',
    'torque_nm',
    'tool_wear_min',
    'Temp_Differential_C',
    'Power_W',
    'Type_Encoded'
]
TARGET_COL = 'machine_failure'

# Handle column name variations
for col in ['Temp_Differential_C', 'Power_W', 'Type_Encoded'
    ]:
    if col not in df.columns:
        lower = col.lower()
        if lower in df.columns:
            df[col] = df[lower]

X = df[FEATURE_COLS].copy()
y = df[TARGET_COL].copy()

print(f"\nClass distribution:")
print(y.value_counts())
print(f"Failure rate: {y.mean()*100:.2f}%")

#
    ============================================================

# STEP 2: Handle Class Imbalance with SMOTE
#
    ============================================================

print("\n" + "=" * 60)
print("STEP 2: Handle Class Imbalance (SMOTE)")
print("=" * 60)

# First split, then apply SMOTE only to training set
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=RANDOM_SEED,
    stratify=y   # maintain class proportions in both sets
)

print(f"Training set: {X_train.shape[0]} samples")
print(f"  - No Failure: {(y_train==0).sum()}")
print(f"  - Failure:    {(y_train==1).sum()}")
print(f"Test set: {X_test.shape[0]} samples")

# Apply SMOTE to training data only
smote = SMOTE(random_state=RANDOM_SEED, k_neighbors=5)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

print(f"\nAfter SMOTE oversampling:")
print(f"  - No Failure: {(y_train_sm==0).sum()}")
print(f"  - Failure:    {(y_train_sm==1).sum()}")
print("  (now balanced!)")

#
    ================================================================

# STEP 3: Feature Scaling (StandardScaler)
#
    ================================================================

print("\n" + "=" * 60)
print("STEP 3: Feature Scaling")
print("=" * 60)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_sm)
X_test_scaled  = scaler.transform(X_test)

# Save scaler for use in Streamlit app
with open(SCALER_OUTPUT, 'wb') as f:
    pickle.dump(scaler, f)
print(f"Scaler saved to: {SCALER_OUTPUT}")

#
    ================================================================

# STEP 4: Train XGBoost Model
#
    ================================================================

print("\n" + "=" * 60)
print("STEP 4: Train XGBoost Classifier")
print("=" * 60)

# XGBoost hyperparameters
xgb_params = {
```

```
    'n_estimators':     300,     # number of trees
    'max_depth':        5,       # maximum tree depth
    'learning_rate':    0.05,    # step size (eta)
    'subsample':        0.8,     # fraction of samples per tree
    'colsample_bytree': 0.8,     # fraction of features per
       tree
    'min_child_weight': 3,       # min samples in leaf node
    'gamma':            0.1,     # min loss reduction to split
    'reg_alpha':        0.1,     # L1 regularization
    'reg_lambda':       1.0,     # L2 regularization
    'random_state':     RANDOM_SEED,
    'eval_metric':      'logloss',
    'use_label_encoder': False
}

print("Hyperparameters:")
for k, v in xgb_params.items():
    print(f"  {k}: {v}")

model = xgb.XGBClassifier(**xgb_params)

# Train with early stopping using eval set
eval_set = [(X_train_scaled, y_train_sm),
            (X_test_scaled, y_test)]

model.fit(
    X_train_scaled, y_train_sm,
    eval_set=eval_set,
    early_stopping_rounds=30,
    verbose=50  # print every 50 trees
)

print(f"\nBest iteration: {model.best_iteration}")

# Save model
with open(MODEL_OUTPUT, 'wb') as f:
    pickle.dump(model, f)
print(f"Model saved to: {MODEL_OUTPUT}")

#
   ============================================================

# STEP 5: Evaluate Model
#
   ============================================================

print("\n" + "=" * 60)
print("STEP 5: Model Evaluation")
print("=" * 60)

# Predictions
```

```
y_pred       = model.predict(X_test_scaled)
y_pred_proba = model.predict_proba(X_test_scaled)[:, 1]

# Classification Report
print("\nClassification Report:")
print(classification_report(y_test, y_pred,
                            target_names=['No Failure', '
                               Failure'],
                            digits=4))

# ROC-AUC
roc_auc = roc_auc_score(y_test, y_pred_proba)
print(f"ROC-AUC Score: {roc_auc:.4f}")

#
    ============================================================

# STEP 6: Plot Results
#
    ============================================================

print("\n" + "=" * 60)
print("STEP 6: Generating Plots")
print("=" * 60)

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# 6a: Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=['No Failure', 'Failure']
)
disp.plot(ax=axes[0], colorbar=False, cmap='Blues')
axes[0].set_title('Confusion Matrix', fontweight='bold')

# 6b: ROC Curve
fpr, tpr, _ = roc_curve(y_test, y_pred_proba)
axes[1].plot(fpr, tpr, color='steelblue', lw=2,
             label=f'AUC = {roc_auc:.4f}')
axes[1].plot([0,1],[0,1], 'k--', lw=1, alpha=0.5)
axes[1].fill_between(fpr, tpr, alpha=0.15, color='steelblue')
axes[1].set_xlabel('False Positive Rate')
axes[1].set_ylabel('True Positive Rate')
axes[1].set_title('ROC Curve', fontweight='bold')
axes[1].legend(loc='lower right')

# 6c: Feature Importance
importances = model.feature_importances_
feat_names  = FEATURE_COLS
sorted_idx  = np.argsort(importances)
```

```
axes[2].barh([feat_names[i] for i in sorted_idx],
             [importances[i] for i in sorted_idx],
             color='steelblue', alpha=0.8)
axes[2].set_xlabel('Feature Importance (gain)')
axes[2].set_title('XGBoost Feature Importance', fontweight='bold')

plt.tight_layout()
fig_path = os.path.join(PLOT_DIR, 'model_evaluation.png')
plt.savefig(fig_path, dpi=150, bbox_inches='tight')
print(f"Saved evaluation plots to: {fig_path}")
plt.show()

# ============================================================

# STEP 7: SHAP Analysis (Model Explainability)
# ============================================================

print("\n" + "=" * 60)
print("STEP 7: SHAP Explainability Analysis")
print("=" * 60)

# SHAP values explain WHY the model made each prediction
explainer   = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test_scaled)

# Summary plot (global feature importance via SHAP)
plt.figure(figsize=(10, 6))
shap.summary_plot(shap_values, X_test,
                  feature_names=FEATURE_COLS,
                  show=False)
plt.title('SHAP Summary Plot - Feature Impact on Failure Prediction',
          fontweight='bold')
plt.tight_layout()
shap_path = os.path.join(PLOT_DIR, 'shap_summary.png')
plt.savefig(shap_path, dpi=150, bbox_inches='tight')
print(f"Saved SHAP plot to: {shap_path}")
plt.show()

# ============================================================

# STEP 8: Cross-Validation
# ============================================================

print("\n" + "=" * 60)
print("STEP 8: Cross-Validation (5-Fold)")
```

```
print("=" * 60)

# Scale the full SMOTE dataset for CV
X_full_scaled = scaler.transform(X_train)  # use original (not SMOTE) for CV
cv_scores = cross_val_score(
    model, X_full_scaled, y_train,
    cv=5, scoring='roc_auc'
)
print(f"5-Fold Cross-Validation ROC-AUC Scores:")
for i, score in enumerate(cv_scores, 1):
    print(f"  Fold {i}: {score:.4f}")
print(f"Mean: {cv_scores.mean():.4f} +/- {cv_scores.std():.4f}")

print("\n=== ML TRAINING COMPLETE ===")
print(f"Model saved: {MODEL_OUTPUT}")
print(f"Plots saved: {PLOT_DIR}")
```

**Listing 6.1.** Complete XGBoost Model Training Script (03_xgboost_model.py)

Run with: `python ml\03_xgboost_model.py`

## 6.4 Evaluation Metrics Explained

**Table 6.1.** Model Evaluation Metrics Definitions

| Metric | What It Measures | Formula |
|---|---|---|
| Accuracy | Overall percentage of correct predictions | $\frac{TP+TN}{TP+TN+FP+FN}$ |
| Precision | Of predicted failures, how many are real | $\frac{TP}{TP+FP}$ |
| Recall | Of actual failures, how many we caught | $\frac{TP}{TP+FN}$ |
| F1-Score | Harmonic mean of Precision and Recall | $\frac{2 \cdot P \cdot R}{P+R}$ |
| ROC-AUC | Area under the ROC curve; overall discrimination | (integral of ROC) |

**Why Recall matters more than Accuracy here:**
In a manufacturing context, failing to detect a machine failure (False Negative) is far more costly than a false alarm (False Positive). A missed failure means unplanned downtime; a false alarm means an unnecessary inspection. Therefore, we prioritize high **Recall** for the failure class, even at some cost to Precision. The XGBoost model achieves this balance effectively.

# Chapter 7

# Experimental Results and Analysis

## 7.1 Expected Experimental Results

After running the training script in Chapter 6, you will obtain results similar to those described below. Your exact numbers will vary slightly due to randomness.

## 7.2 Model Performance Summary

**Table 7.1.** XGBoost Model Performance on Test Set (Expected Range)

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| No Failure (0) | 0.98-0.99 | 0.97-0.99 | 0.98-0.99 | ∼1932 |
| Failure (1) | 0.75-0.88 | 0.80-0.92 | 0.78-0.90 | ∼68 |
| **Weighted Avg** | 0.97-0.98 | 0.97-0.98 | 0.97-0.98 | 2000 |

**Table 7.2.** Overall Model Metrics

| Metric | Expected Value |
|---|---|
| Overall Accuracy | 97-98% |
| ROC-AUC Score | 0.95-0.98 |
| 5-Fold CV Mean AUC | 0.93-0.97 |
| Best Iteration (trees) | 150-250 |
| Training Time | 5-15 seconds |

## 7.3 Confusion Matrix Analysis

The confusion matrix will show four quadrants:

- **True Negatives (TN)**: Operations correctly predicted as normal. Expected: ∼1900. These are the majority class.
- **False Positives (FP)**: Normal operations incorrectly flagged as failures. Expected: ∼30-50. Causes unnecessary maintenance checks-an acceptable cost.

- **False Negatives (FN)**: Actual failures that the model missed. Expected: ∼5-15. These are the critical errors to minimize.
- **True Positives (TP)**: Failures correctly predicted. Expected: ∼55-65.

The model correctly catches approximately **85-92% of all actual failures** (high recall for the failure class).

## 7.4 Feature Importance Analysis

Based on XGBoost's built-in feature importance (gain metric), the expected ranking is:

**Table 7.3.** Feature Importance Ranking from XGBoost

| Rank | Feature | Relative Importance | Engineering Insight |
|---|---|---|---|
| 1 | `torque_nm` | Very High | High torque → mechanical stress |
| 2 | `tool_wear_min` | High | Worn tools fail more |
| 3 | `Power_W` | High | Power spikes indicate instability |
| 4 | `rotational_speed_rpm` | Medium | Speed deviations are anomalies |
| 5 | `Temp_Differential_C` | Medium | Thermal stress indicator |
| 6 | `process_temp_k` | Low-Medium | Direct temperature reading |
| 7 | `air_temp_k` | Low | Ambient temperature |
| 8 | `Type_Encoded` | Low | Product type has some influence |

## 7.5 SHAP Analysis Results

The SHAP (SHapley Additive exPlanations) analysis provides model interpretability. Key findings from the SHAP summary plot:

- **High torque** (red dots on the right) consistently pushes the prediction toward failure. Specifically, torque $> 60$ Nm significantly increases failure probability.
- **High tool wear** (red dots on the right) increases failure probability. Operations with tool wear $> 200$ minutes have noticeably higher failure risk.
- **High rotational speed** (red dots on the left) actually *reduces* failure probability-consistent with the negative correlation found during EDA. Failures tend to occur at low RPM settings.
- **Large temperature differential** increases failure risk, confirming the domain knowledge that thermal management is critical.

These findings are directly actionable: plant engineers should monitor torque levels and proactively replace tools before reaching 200 minutes of wear.

## 7.6 Cross-Validation Results Discussion

The 5-fold cross-validation with mean AUC $\approx 0.95$ (+/-0.02) shows that the model generalizes well across different subsets of the data. The low standard deviation (+/-

0.02) indicates the model is stable and not overfitting to a particular subset of training examples.

# Chapter 8

# Knowledge Presentation: Power BI Dashboard

## 8.1 Connecting Power BI to PostgreSQL

**Connect Power BI to Your DW**

1. Open **Power BI Desktop**.
2. Click **Home** tab → **Get Data** → **PostgreSQL database**.
3. Server: `localhost`
4. Database: `co4031_dw`
5. Click **OK**.
6. When prompted for credentials, select **Database** tab, enter username `postgres` and your password.
7. Click **Connect**.
8. In the Navigator, expand **dw** schema and select:
   - `dw.fact_machine_operations`
   - `dw.dim_product`
   - `dw.dim_failure_type`
   - `dw.dim_machine_condition`
   - `dw.dim_date`
9. Click **Load**.

## 8.2 Create Relationships Between Tables

**Define Star Schema Relationships in Power BI**

1. Go to the **Model** view (icon on left panel showing a grid/network).
2. You should see all 5 tables. If relationships are not auto-detected:
3. Drag `fact_machine_operations.product_fk` to `dim_product.product_sk`.
4. Drag `fact_machine_operations.failure_type_fk` to `dim_failure_type.failure_type_sk`.
5. Drag `fact_machine_operations.condition_fk` to `dim_machine_condition.condition_sk`.

6. Drag `fact_machine_operations.date_fk` to `dim_date.date_sk`.
7. All relationships should be Many-to-One (arrow pointing from fact to dim).

## 8.3 Create DAX Measures

In Power BI, click **New Measure** (in the Home tab or right-click a table) and enter each of these DAX formulas:

```
-- Measure 1: Total Operations
Total Operations =
COUNTROWS('dw fact_machine_operations')

-- Measure 2: Total Failures
Total Failures =
CALCULATE(
    COUNTROWS('dw fact_machine_operations'),
    'dw fact_machine_operations'[machine_failure] = 1
)

-- Measure 3: Failure Rate (%)
Failure Rate % =
DIVIDE([Total Failures], [Total Operations], 0) * 100

-- Measure 4: Average Torque
Avg Torque (Nm) =
AVERAGE('dw fact_machine_operations'[torque_nm])

-- Measure 5: Average Tool Wear
Avg Tool Wear (min) =
AVERAGE('dw fact_machine_operations'[tool_wear_min])

-- Measure 6: Avg Power (W)
Avg Power (W) =
AVERAGE('dw fact_machine_operations'[power_w])
```

**Listing 8.1.** Power BI DAX Measures

## 8.4 Dashboard Page 1: Executive Overview

Build the following visuals on the first page ("Executive Overview"):

1. **KPI Cards** (top row):
   - Card: "Total Operations" using measure `[Total Operations]`
   - Card: "Total Failures" using measure `[Total Failures]`
   - Card: "Overall Failure Rate" using measure `[Failure Rate %]`
   - Card: "Avg Tool Wear" using measure `[Avg Tool Wear (min)]`

2. **Bar Chart**: Failures by Product Quality Type
   - X-axis: `dim_product.quality_label`
   - Y-axis: `[Total Failures]`
   - Color: by quality type
3. **Donut Chart**: Failure Distribution by Type
   - Legend: `dim_failure_type.failure_name`
   - Values: `[Total Failures]`
4. **Line Chart**: Monthly Failure Trend
   - X-axis: `dim_date.month_name`
   - Y-axis: `[Total Failures]`
5. **Slicer**: Filter by Quality Type
   - Field: `dim_product.quality_label`
   - Style: Dropdown

## 8.5 Dashboard Page 2: Sensor Analysis

1. **Scatter Chart**: Torque vs Rotational Speed (colored by failure)
   - X-axis: `torque_nm`
   - Y-axis: `rotational_speed_rpm`
   - Play Axis: `month_name` (animated over time)
   - Color: `machine_failure`
2. **Histogram** (use Column Chart with bin ranges): Distribution of Tool Wear values at failure vs non-failure
3. **Gauge Chart**: Average Tool Wear
   - Value: `[Avg Tool Wear (min)]`
   - Max: 250 (the critical threshold)
   - Target: 200 (the warning threshold)
4. **Matrix Table**: Machine Condition vs Failure Rate
   - Rows: `dim_machine_condition.condition_label`
   - Values: `[Total Operations]`, `[Total Failures]`, `[Failure Rate %]`

## 8.6 Formatting and Publishing

**Format and Save Your Dashboard**

1. Add a title text box to each page with the page name.
2. Set a consistent color theme: **Format** → **Themes** → choose "Executive" or a corporate blue theme.
3. Add your team name and date as a footer text box.
4. Save as: `C:\co4031_project\dashboard\co4031_dashboard.pbix`
5. Export screenshots: **File** → **Export** → **Export to PDF** for your report.

# Chapter 9

# Knowledge Application: Streamlit Web App

## 9.1 What the Streamlit App Does

The Streamlit web application serves as a practical deployment of the XGBoost model. An operator at a manufacturing plant can open the web app, enter the current machine's sensor readings, and immediately receive:

- A **failure probability score** (0% to 100%).
- A **risk level** (Low / Medium / High / Critical).
- A **recommended action** for the operator.
- A visualization of which sensor values are most contributing to the risk.

## 9.2 Build the Streamlit Application

Create: `C:\co4031_project\streamlit_app\app.py`

```
"""
CO4031 BTL - Streamlit Web App: Machine Failure Prediction
Run with: streamlit run app.py
"""

import streamlit as st
import pandas as pd
import numpy as np
import pickle
import plotly.graph_objects as go
import plotly.express as px
import sys
import os

# Add ml directory to path so we can load model
sys.path.append(r"C:\co4031_project\ml")

#
```

```
    # ============================================================

# PAGE CONFIGURATION
#
    # ============================================================

st.set_page_config(
    page_title="Machine Failure Prediction System",
    layout="wide",
    initial_sidebar_state="expanded"
)

# Custom CSS for better appearance
st.markdown("""
<style>
    .main-header {
        background: linear-gradient(135deg, #1e3a5f 0%, #2980
           b9 100%);
        color: white;
        padding: 20px;
        border-radius: 10px;
        margin-bottom: 20px;
        text-align: center;
    }
    .metric-card {
        background: white;
        border: 1px solid #e0e0e0;
        border-radius: 8px;
        padding: 15px;
        text-align: center;
        box-shadow: 2px 2px 5px rgba(0,0,0,0.05);
    }
    .risk-low      { background-color: #d5f5e3; border-left:
       5px solid #27ae60; }
    .risk-medium   { background-color: #fef9e7; border-left:
       5px solid #f1c40f; }
    .risk-high     { background-color: #fde8e8; border-left:
       5px solid #e74c3c; }
    .risk-critical { background-color: #f0d0d0; border-left:
       5px solid #922b21; }
</style>
""", unsafe_allow_html=True)

#
    # ============================================================

# LOAD MODEL AND SCALER
#
    # ============================================================

@st.cache_resource  # Cache so it only loads once
```

```python
def load_model():
    model_path  = r"C:\co4031_project\ml\xgboost_model.pkl"
    scaler_path = r"C:\co4031_project\ml\scaler.pkl"
    try:
        with open(model_path, 'rb') as f:
            model = pickle.load(f)
        with open(scaler_path, 'rb') as f:
            scaler = pickle.load(f)
        return model, scaler, True
    except FileNotFoundError:
        return None, None, False

model, scaler, model_loaded = load_model()

# ============================================================

# HEADER
# ============================================================

st.markdown("""
<div class="main-header">
    <h1>Manufacturing Equipment Failure Prediction System</h1>
    <p>CO4031 Data Warehouse | Real-time Machine Health Assessment</p>
</div>
""", unsafe_allow_html=True)

if not model_loaded:
    st.error("[ERROR] Model not found! Please run the training script first: "
             "python ml/03_xgboost_model.py")
    st.stop()

# ============================================================

# SIDEBAR: Input Parameters
# ============================================================

st.sidebar.header("[Settings] Machine Parameters")
st.sidebar.markdown("Enter current sensor readings:")

# Product Type
quality_type = st.sidebar.selectbox(
    "Product Quality Type",
    options=["Low (L)", "Medium (M)", "High (H)"],
    index=1
```

```
)
type_encoded = {"Low (L)": 0, "Medium (M)": 1, "High (H)":
    2}[quality_type]

# Temperature inputs
st.sidebar.subheader("[Sensors] Temperature")
air_temp_c = st.sidebar.slider(
    "Air Temperature (C)",
    min_value=15.0, max_value=35.0,
    value=25.0, step=0.1
)
process_temp_c = st.sidebar.slider(
    "Process Temperature (C)",
    min_value=30.0, max_value=70.0,
    value=55.0, step=0.1
)

# Convert to Kelvin for model
air_temp_k     = air_temp_c + 273.15
process_temp_k = process_temp_c + 273.15
temp_diff      = process_temp_c - air_temp_c

# Mechanical sensors
st.sidebar.subheader("[Sensors] Mechanical")
rotational_speed = st.sidebar.slider(
    "Rotational Speed (RPM)",
    min_value=1000, max_value=3000,
    value=1500, step=10
)
torque_nm = st.sidebar.slider(
    "Torque (Nm)",
    min_value=0.0, max_value=100.0,
    value=40.0, step=0.5
)
tool_wear_min = st.sidebar.slider(
    "Tool Wear Duration (minutes)",
    min_value=0, max_value=280,
    value=100, step=1
)

# Derived feature
omega_rad_s = (2 * np.pi * rotational_speed) / 60
power_w     = torque_nm * omega_rad_s

#
    ============================================================

# PREDICTION
#
    ============================================================
```

```
feature_values = np.array([[
    air_temp_k, process_temp_k, rotational_speed,
    torque_nm, tool_wear_min, temp_diff, power_w,
       type_encoded
]])

feature_scaled = scaler.transform(feature_values)
failure_proba  = model.predict_proba(feature_scaled)[0, 1]
failure_pred   = model.predict(feature_scaled)[0]
failure_pct    = failure_proba * 100

# Risk level classification
if failure_pct < 20:
    risk_level = "LOW"
    risk_color = "#27ae60"
    risk_emoji = "[OK]"
    action = ("Machine is operating normally. "
              "Continue standard monitoring schedule.")
    css_class = "risk-low"
elif failure_pct < 50:
    risk_level = "MEDIUM"
    risk_color = "#f1c40f"
    risk_emoji = "[WARN]"
    action = ("Elevated risk detected. Schedule inspection "
              "within the next 24 hours.")
    css_class = "risk-medium"
elif failure_pct < 80:
    risk_level = "HIGH"
    risk_color = "#e74c3c"
    risk_emoji = "[HIGH]"
    action = ("High failure risk! Reduce workload immediately
        "
              "and schedule urgent maintenance.")
    css_class = "risk-high"
else:
    risk_level = "CRITICAL"
    risk_color = "#922b21"
    risk_emoji = "[CRIT]"
    action = ("CRITICAL: Stop machine operation immediately.
       "
              "Perform emergency maintenance before
                 restarting.")
    css_class = "risk-critical"

#
    ============================================================

# MAIN DASHBOARD: Results
#
    ============================================================
```

```
col1, col2, col3 = st.columns([2, 1, 1])

with col1:
    # Gauge chart for failure probability
    fig_gauge = go.Figure(go.Indicator(
        mode="gauge+number+delta",
        value=failure_pct,
        domain={'x': [0, 1], 'y': [0, 1]},
        title={'text': "Failure Probability (%)", 'font': {'
            size': 18}},
        gauge={
            'axis': {'range': [0, 100], 'tickwidth': 1},
            'bar':  {'color': risk_color, 'thickness': 0.3},
            'steps': [
                {'range': [0,  20], 'color': '#d5f5e3'},
                {'range': [20, 50], 'color': '#fef9e7'},
                {'range': [50, 80], 'color': '#fde8e8'},
                {'range': [80,100], 'color': '#fadbd8'}
            ],
            'threshold': {
                'line': {'color': "red", 'width': 4},
                'thickness': 0.75,
                'value': 50
            }
        }
    ))
    fig_gauge.update_layout(height=300, margin=dict(t=40, b
        =20, l=20, r=20))
    st.plotly_chart(fig_gauge, use_container_width=True)

with col2:
    st.metric("Risk Level", f"{risk_emoji} {risk_level}")
    st.metric("Failure Probability", f"{failure_pct:.1f}%")
    st.metric("Model Prediction",
              "[FAILED]" if failure_pred == 1 else "[NORMAL]"
                 )

with col3:
    st.metric("Tool Wear Status",
              "Replace Now!" if tool_wear_min > 240
              else "Monitor" if tool_wear_min > 200
              else "OK")
    st.metric("Calculated Power", f"{power_w:.0f} W")
    st.metric("Temp Differential", f"{temp_diff:.1f} C")

# Risk Alert Box
st.markdown(f"""
<div style="padding:15px; border-radius:8px; margin:10px 0;
            background-color:{'#fde8e8' if failure_pct > 50
               else '#d5f5e3'};
            border-left: 6px solid {risk_color};">
```

```
    <h3>{risk_emoji} Risk Level: {risk_level}</h3>
    <p>{action}</p>
</div>
""", unsafe_allow_html=True)

#
    ============================================================

# SENSOR READINGS DISPLAY
#
    ============================================================

st.subheader("Current Sensor Readings")

sensor_data = {
    "Parameter": [
        "Air Temperature", "Process Temperature", "Temp
           Differential",
        "Rotational Speed", "Torque", "Tool Wear", "Power
           Output"
    ],
    "Value": [
        f"{air_temp_c:.1f} C", f"{process_temp_c:.1f} C",
        f"{temp_diff:.1f} C",
        f"{rotational_speed} RPM",
        f"{torque_nm:.1f} Nm",
        f"{tool_wear_min} min",
        f"{power_w:.0f} W"
    ],
    "Status": [
        "Normal" if 20 <= air_temp_c <= 30 else "Elevated",
        "Normal" if process_temp_c <= 60 else "High",
        "Normal" if temp_diff <= 12 else "Elevated",
        "Normal" if 1000 <= rotational_speed <= 2500 else "
           Check",
        "Normal" if torque_nm <= 55 else "High",
        ("New" if tool_wear_min <= 100
         else "Mid" if tool_wear_min <= 200
         else "Worn"),
        "Normal"
    ]
}
st.dataframe(
    pd.DataFrame(sensor_data),
    use_container_width=True,
    hide_index=True
)

#
    ============================================================
```

```python
# FEATURE IMPORTANCE BAR
#    ============================================================

st.subheader("Key Risk Factors")

feature_names  = [
    'Air Temp (K)', 'Process Temp (K)', 'Speed (RPM)',
    'Torque (Nm)', 'Tool Wear (min)', 'Temp Diff (C)',
    'Power (W)', 'Product Type'
]
importances    = model.feature_importances_
feat_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': importances
}).sort_values('Importance', ascending=True)

fig_imp = px.bar(
    feat_df, x='Importance', y='Feature',
    orientation='h',
    title='Feature Importance (How Much Each Factor
       Influences Prediction)',
    color='Importance',
    color_continuous_scale='RdYlGn_r'
)
fig_imp.update_layout(height=350, showlegend=False)
st.plotly_chart(fig_imp, use_container_width=True)

#    ============================================================

# FOOTER
#    ============================================================

st.markdown("---")
st.markdown(
    "**CO4031 Data Warehouse Project** | "
    "Manufacturing Equipment Failure Prediction | "
    "Model: XGBoost Classifier | "
    "Dashboard: Streamlit + Plotly"
)
```

**Listing 9.1.** Complete Streamlit Web Application (app.py)

## 9.3 Running the Streamlit App

**Launch the Web Application**

1. Open Command Prompt.
2. Navigate: `cd C:\co4031_project\streamlit_app`
3. Run: `streamlit run app.py`
4. Your browser will automatically open at `http://localhost:8501`
5. Use the sidebar sliders to input sensor values and watch the prediction update in real time.

**For the Report**

Take a screenshot of the Streamlit app showing:

- A LOW risk scenario (normal operating conditions)
- A HIGH risk scenario (set torque $>$ 70 Nm and tool wear $>$ 240 min)

Include both screenshots in Section 5 of your report.

# Chapter 10

# Conclusion and Future Work

## 10.1 Summary of Main Findings

This project successfully designed and implemented a complete Data Warehouse system for manufacturing equipment failure prediction. The following major findings and contributions were achieved:

### 10.1.1 Technical Contributions

1. **ETL Pipeline**: A fully automated ETL pipeline was implemented using both Python (for scripting and reusability) and Pentaho PDI (for visual process representation). The pipeline performs:
   - Data extraction from CSV source files
   - 12 distinct transformation operations including unit conversion, feature engineering, business logic application, and encoding
   - Batch loading of ∼10,000 records to PostgreSQL in under 3 seconds
2. **Star Schema Data Warehouse**: A well-structured Star Schema was designed with 1 fact table and 4 dimension tables, enabling efficient multi-dimensional analysis of machine failure data across time, product type, failure type, and machine condition dimensions.
3. **OLAP Analytics**: Four OLAP queries demonstrated the analytical power of the Data Warehouse, revealing that:
   - Medium-quality (M-type) products have the highest absolute failure count
   - Tool Wear Failure (TWF) and Overstrain Failure (OSF) are the most common
   - Machines in the "Worn Tool" condition have >10× the failure rate of new tools
4. **Machine Learning Model**: The XGBoost classifier achieved 97-98% accuracy and 0.95-0.98 ROC-AUC, with high recall for the failure class (>85%). The model correctly identifies the vast majority of impending failures, enabling proactive maintenance.
5. **Operational Dashboard**: The Power BI dashboard provides plant managers

with real-time visibility into equipment health KPIs, failure trends, and operational patterns.

6. **Web Application**: The Streamlit app provides a user-friendly interface for operators to perform real-time failure risk assessment without any data science knowledge.

### 10.1.2 Domain Insights

From the data analysis, three key engineering insights emerged:

- **Torque is the primary failure predictor**: Operations with torque $> 60$ Nm are significantly more likely to result in failure. This suggests that torque limits should be enforced at the control system level.
- **Tool replacement at 200 minutes (not 240)**: The data shows failure rates increase dramatically between 200 and 240 minutes of tool wear. Current maintenance schedules should be revised to replace tools at 200 minutes rather than waiting until 240 minutes.
- **Thermal management is underappreciated**: The temperature differential feature (a derived variable not in the original sensor suite) proved to be a significant predictor. Adding dedicated thermal monitoring sensors would improve predictive capability.

## 10.2 Limitations of the Current Study

1. **Synthetic dataset**: The AI4I 2020 dataset is realistically modeled but synthetic. Performance on real factory data may differ due to sensor noise, hardware variation, and unmeasured variables.
2. **No real-time ingestion**: The current pipeline is batch-based (processes historical data). A production system would require streaming ingestion (e.g., Apache Kafka + Spark Streaming) to process live sensor data.
3. **No temporal modeling**: XGBoost treats each operation as independent. In reality, failure risk builds up over time. An LSTM or Transformer model could capture these temporal patterns.
4. **Class imbalance**: Despite SMOTE, the severe class imbalance (3.5% failures) means the model's failure-class precision is moderate ($\sim$75-88%). More failure data would improve precision.
5. **Single machine type**: The dataset represents a single type of machine. A real factory has many machine types, each requiring its own tailored model.

## 10.3 Future Research Directions

1. **Streaming Data Pipeline**: Implement Apache Kafka for real-time sensor data ingestion. This would transform the system from reactive analysis to true real-time monitoring. Tools: Kafka + Spark Streaming + TimescaleDB.

2. **Advanced ML Models**: Explore LSTM networks for time-series failure prediction, capturing the temporal dependency between consecutive operations.
3. **Cloud Deployment**: Migrate the entire stack to AWS:
   - Amazon S3 for raw data storage
   - AWS Glue for managed ETL
   - Amazon Redshift for cloud Data Warehouse
   - Amazon SageMaker for model deployment
   - Amazon QuickSight for dashboards
4. **Anomaly Detection**: Add unsupervised anomaly detection (e.g., Isolation Forest, Autoencoder) to detect unusual sensor patterns that may not match known failure types.
5. **Multi-machine Extension**: Extend the Data Warehouse schema to support multiple machine types, production lines, and factory sites using a more complex Snowflake Schema.
6. **Mobile Application**: Develop a mobile companion app for the Streamlit dashboard so maintenance engineers can check machine health from anywhere on the factory floor.

# Chapter 11

# References


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# Appendix A

# Troubleshooting Common Problems

## A.1 Python Issues

**Table A.1.** Common Python Error Solutions

| Error Message | Solution |
|---|---|
| `ModuleNotFoundError: No module named 'xgboost'` | Run: `pip install xgboost` |
| `No such file or directory: 'ai4i2020.csv'` | Make sure the CSV is in the exact path shown. Check for typos. |
| `OperationalError: could not connect to server` | Make sure PostgreSQL service is running: open Services (Win+R, type `services.msc`), find "PostgreSQL 15", click Start. |
| `UnicodeDecodeError` | Add `encoding='utf-8'` to `pd.read_csv()`. |
| `pip: command not found` | Python was not added to PATH. Reinstall Python with "Add to PATH" checkbox checked. |

## A.2 PostgreSQL Issues

1. **Cannot connect with pgAdmin**:
   - Check PostgreSQL service is running (see above).
   - Try password `postgres` without numbers first.
   - Reinstall PostgreSQL if password is completely forgotten.
2. **SQL error: relation does not exist**: This means you are querying the wrong schema. Make sure you run `CREATE SCHEMA IF NOT EXISTS dw;` first.

3. **Permission denied**: Make sure you are logged in as the `postgres` user in pgAdmin.

## A.3 Power BI Issues

1. **Cannot connect to PostgreSQL**: You need to install the PostgreSQL ODBC driver: https://www.postgresql.org/ftp/odbc/versions/msi/ Download `psqlodbc_15_00_0000.zip`, extract, and run the installer.
2. **Relationships not working**: Delete all existing relationships and recreate them manually in Model view.
3. **Measure shows error**: Check that all column references in DAX match the exact column names (case-sensitive in some versions).

## A.4 Pentaho PDI Issues

1. **Spoon.bat does not open**: Check that Java 11 is installed. Set `JAVA_HOME` environment variable:
   - Open System Properties → Environment Variables.
   - New System Variable: Name=`JAVA_HOME`, Value=`C:\Program Files\Java\jdk-11.x.x`
   - Add `%JAVA_HOME%\bin` to the PATH variable.
2. **Database connection fails**: Verify the PostgreSQL JDBC driver (`.jar` file) is in `pentaho\data-integration\lib\` and that Spoon was restarted after adding it.

# Appendix B

# Recommended Project Timeline

**Table B.1.** 3-Week Project Schedule

| Week | Days | Tasks | Owner |
|---|---|---|---|
| Week 1 | Day 1 | Install all software (Chapter 0) | All |
| | Day 1-2 | Download dataset; run preprocessing script | Member 1 |
| | Day 2-3 | Set up PostgreSQL; create staging tables | Member 2 |
| | Day 3-4 | Build Pentaho ETL transformation | Member 1 |
| | Day 4-5 | Run Python load script; verify data in DB | All |
| Week 2 | Day 6-7 | Create Star Schema DDL; populate dimensions | Member 2 |
| | Day 7-8 | Run OLAP queries; verify results | Member 2 |
| | Day 8-9 | Train XGBoost model; generate plots | Member 3 |
| | Day 9-10 | Perform SHAP analysis; interpret results | Member 3 |
| | Day 10 | Team review: check all outputs are correct | All |
| Week 3 | Day 11-12 | Build Power BI dashboard (2 pages) | Member 2 |
| | Day 12-13 | Build Streamlit web application | Member 3 |
| | Day 13-14 | Write report sections 1-7 | All |
| | Day 14 | Final review; submit Drive link; prepare presentation | All |

# Appendix C

# Final Report and Submission Checklist

Use this checklist before submitting. Every item should be checked off.

## C.1 Technical Deliverables

- ☐ Raw dataset (`ai4i2020.csv`) is in `data/raw/`
- ☐ Cleaned dataset (`machine_data_cleaned.csv`) is in `data/cleaned/`
- ☐ EDA plots (`eda_distributions.png`, `correlation_heatmap.png`) are saved
- ☐ Pentaho transformation (`machine_etl.ktr`) is saved and runs without errors
- ☐ PostgreSQL DW is created with all 5 tables (1 fact + 4 dims)
- ☐ All 4 OLAP queries run successfully and return correct results
- ☐ XGBoost model (`xgboost_model.pkl`) is saved
- ☐ Model achieves >90% accuracy and >0.90 ROC-AUC
- ☐ All evaluation plots are saved in `ml/plots/`
- ☐ Power BI dashboard has at least 2 pages with 8+ visuals
- ☐ Streamlit app runs without errors at `localhost:8501`
- ☐ All screenshots are taken for the report

## C.2 Report Sections (7 Required)

- ☐ Section 1: Introduction and Statement of Purpose (2-3 pages)
- ☐ Section 2: Data Preprocessing (3-4 pages + screenshots)
- ☐ Section 3: Data Mining Methodology (3-4 pages)
- ☐ Section 4: Experimental Results and Analysis (3-4 pages + tables/charts)
- ☐ Section 5: Knowledge Presentation / Application (2-3 pages + screenshots)

☐ Section 6: Conclusion and Future Work (2 pages)

☐ Section 7: References (at least 10 references)

## C.3 Submission Requirements

☐ Drive link is submitted at least 15 minutes before your group's turn

☐ All team members are present for the presentation

☐ Each team member can explain their assigned components

☐ Report is formatted in A4, 12pt font, with page numbers

☐ Report has a title page with all member names and student IDs

## C.4 Grading Rubric Summary

**Table C.1.** Expected Grading Criteria

| Component | Key Requirements | Weight |
|---|---|---|
| ETL Pipeline | Extraction + Transformation (cleaning, derived features) + Load. Pentaho visualization. | 25% |
| Data Warehouse | Star schema with fact + dim tables. OLAP queries. | 25% |
| Dashboard | Power BI with multiple page types, KPIs, trend charts. | 20% |
| AI / ML | XGBoost or equivalent. Metrics reported. Feature importance. | 20% |
| Report Quality | All 7 sections present. Clear writing. Proper references. | 10% |

# Appendix D

# Glossary of Technical Terms

**AUC** Area Under the (ROC) Curve. A single number (0-1) measuring how well a classifier ranks positive examples above negative ones. AUC = 1.0 is perfect; AUC = 0.5 is random guessing.

**Class Imbalance** When one class (e.g., "No Failure" = 96.5%) vastly outnumbers another (e.g., "Failure" = 3.5%) in the training dataset. Can cause the model to ignore the minority class.

**CSV** Comma-Separated Values. A plain text file format where each row is a record and columns are separated by commas.

**DAX** Data Analysis Expressions. The formula language used in Power BI to create calculated columns and measures.

**Dimension Table** In a Star Schema, a table containing descriptive attributes (who, what, when) that provide context to the facts.

**ETL** Extract, Transform, Load. The three-phase process of moving data from source systems to a Data Warehouse.

**F1-Score** The harmonic mean of Precision and Recall. Useful when you need a balance between both metrics.

**Fact Table** The central table in a Star Schema. Contains numerical measurements and foreign keys to dimension tables.

**Feature Engineering** The process of creating new input variables (features) from existing raw data to improve model performance.

**Foreign Key** A column in one table that references the primary key of another table, establishing a relationship between them.

**IQR** Interquartile Range. The difference between the 75th and 25th percentiles of a dataset. Used to detect outliers.

**JDBC** Java Database Connectivity. A standard interface for Java programs to connect to databases (used by Pentaho).

**KPI** Key Performance Indicator. A measurable value that demonstrates how effectively an objective is being achieved.

**OLAP** Online Analytical Processing. Queries that aggregate data across

multiple dimensions for business intelligence analysis.

**SMOTE** Synthetic Minority Over-sampling Technique. An algorithm that creates synthetic examples of the minority class to balance the dataset.

**Staging Table** A temporary database table used as an intermediate storage area during ETL, before data is loaded into the final DW tables.

**Star Schema** A Data Warehouse design pattern with a central fact table surrounded by dimension tables, resembling a star shape.

**XGBoost** Extreme Gradient Boosting. A tree ensemble machine learning algorithm known for its speed and performance on tabular data.